

University of Information Technology & Sciences (UITs)

Faculty of Science and Engineering

Department of Computer Science and Engineering

Program: B.Sc. in CSE

Term Final Examination, Spring 2024

Course Title: Electronic Device and Circuits

Course Code: EEE 251

Marks: 50

Time: 3(three) hours

(Answer all questions)

1. a) Describe the operating principle of a pnp bipolar junction transistor with necessary diagrams. [05]
- b) Find the range of R_L and I_L that will keep the zener diode on for the circuit shown in Figure 1. Also find maximum power rating. [05]

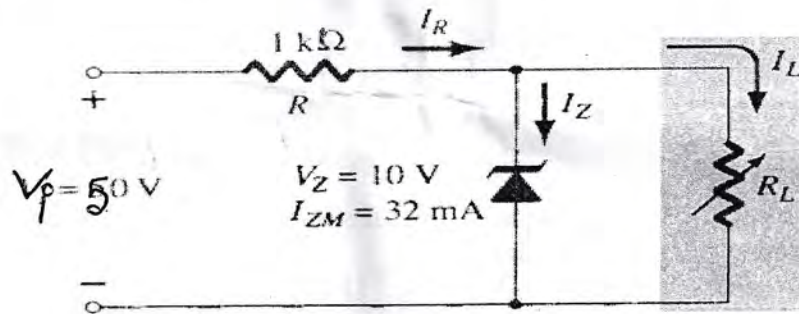


Figure 1

2. a) Describe the characteristics and operating principle of E-MOSFET with necessary diagrams. Also explain CMOS Inverter. [06]
- b) Design the fixed bias circuit shown in Figure 2. Load line is given. [04]

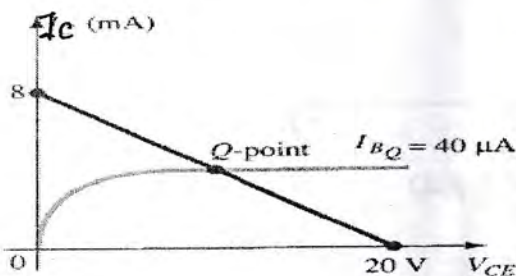
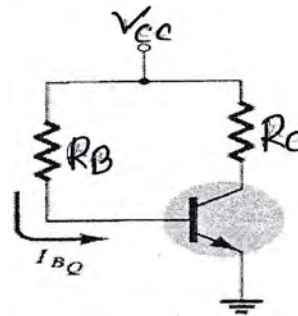


Figure 2



3. a) Find the output voltage for the circuit shown in Figure 3.

[04]

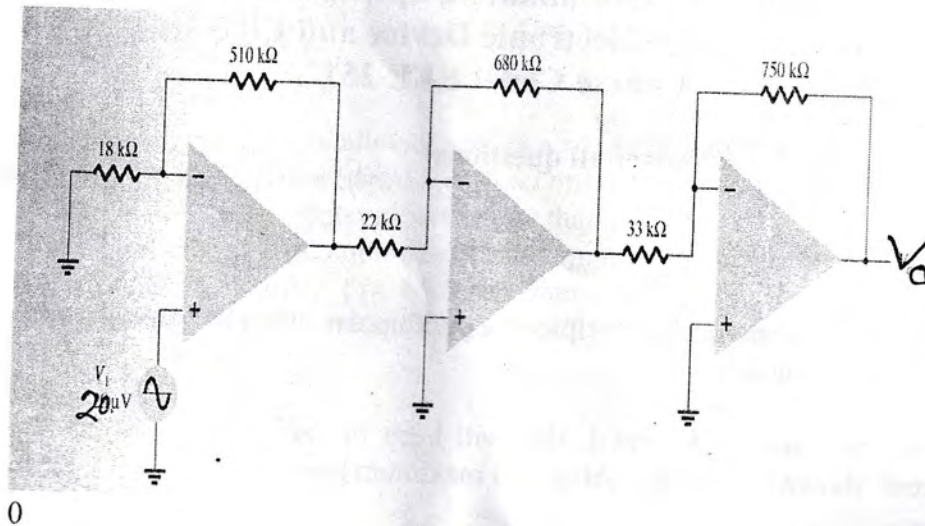


Figure 3

- b) For the emitter bias configuration circuit shown in Figure 4, determine the labeled currents and voltages.

[06]

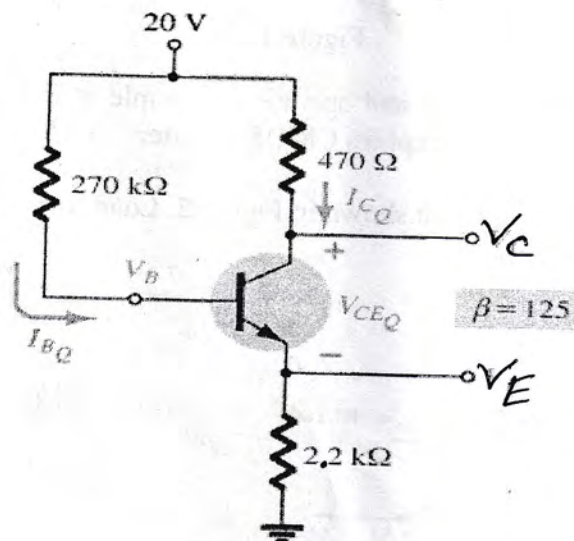


Figure 4

4. a) Design the circuit shown in Figure 5. Let $V_t = 1\text{ V}$ and $k'_n(W/L) = 1\text{ mA/V}^2$

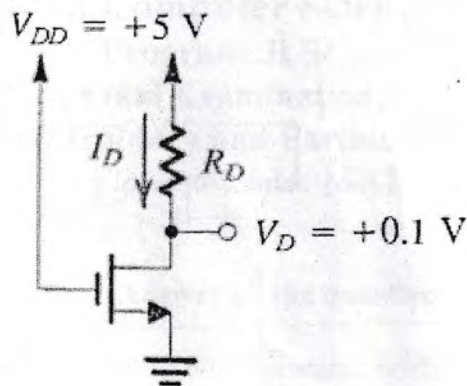


Figure 5

- b) Derive the equation of output voltage of non inverting amplifier and integrator with necessary circuit diagrams. [05]
5. a) For the circuit shown in Figure 6, determine the labeled node voltages. [05]
 $V_t = 1\text{ V}$ and $k'_n(W/L) = 2\text{ mA/V}^2$

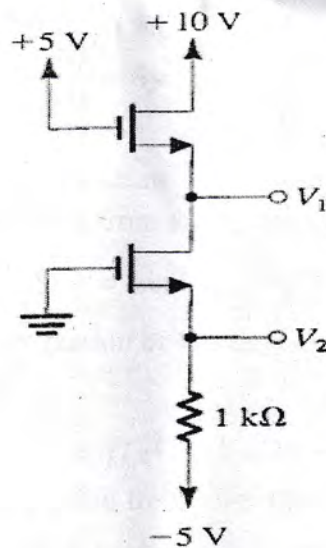


Figure 6

- b) Sketch the output voltage (V_o) of circuit shown in Figure 7 with necessary explanation. Input voltage is given. [05]

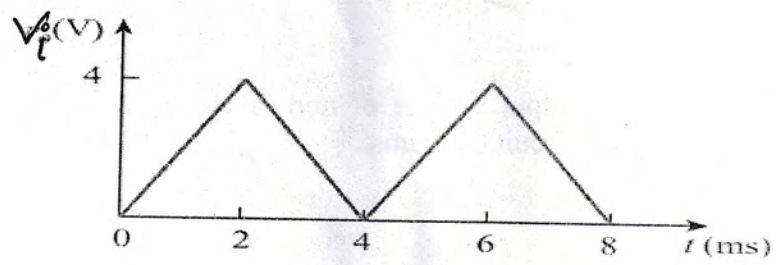
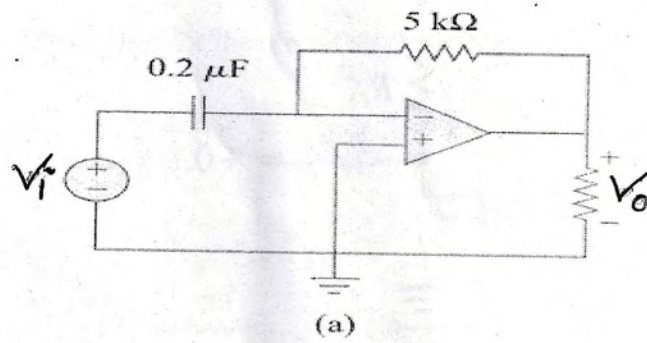


Figure 7